Tactile Sensors Are an Effective Substitute for Wrist Cameras in Contact-Rich or Occluded Manipulation

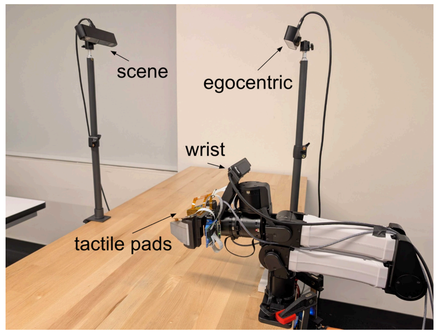
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Toyota Research Institute

1. Motivation

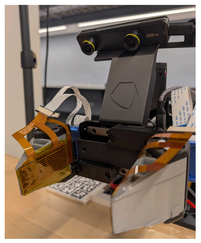
- Wrist cameras may miss finger–object contact due to occlusion.
- Finger-mounted tactile sensors directly measure local contact.

Q: **Can tactile sensing substitute for wrist-camera feedback in learned robotic manipulation?**

2. Methodology



- 2 FlexiTac sensors
- YAM robot arm
- Raiden data collection toolkit

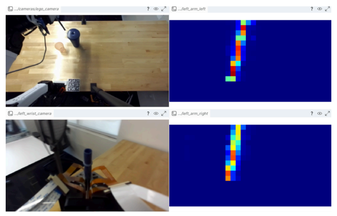


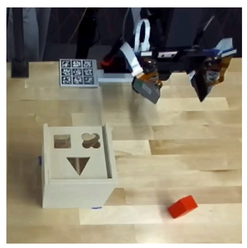
Hardware and Sensing

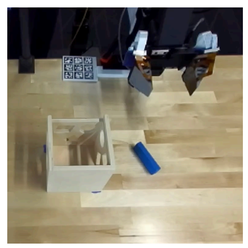
- Scene and egocentric RGB → visual context
- A wrist-mounted RGB camera is optionally enabled
- Tactile maps directly encode local finger–object contact

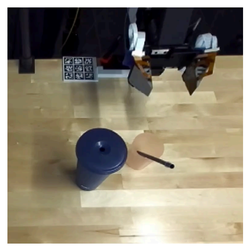
Six Real-World Manipulation Tasks

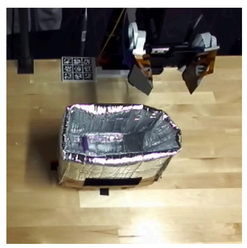
T1 Tight-clearance placement
T2 Block / sponge placement
T3 Straw insertion (shown on the right)
T4 Occluded retrieval
T5 Plug insertion
T6 Whiteboard wiping




T1 (tight clearance)


T2 (block/sponge)


T3 (insertion)


T4 (blind retrieval)


T5 (occluded insertion)


T6 (sustained contact)

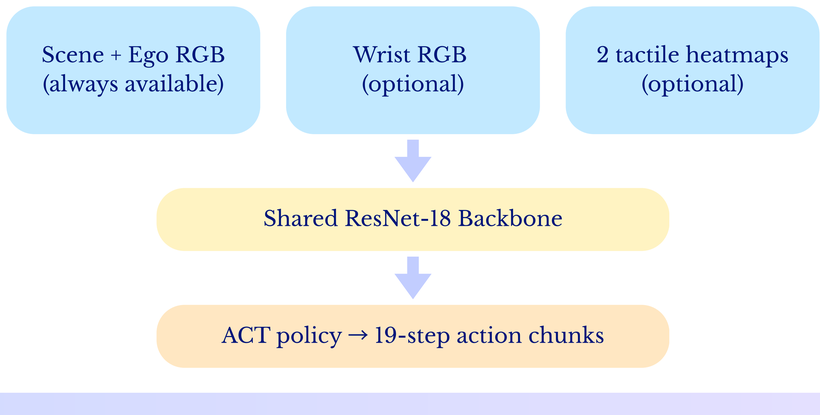
Visuotactile ACT Policy

- Shared ImageNet-pretrained ResNet-18 encodes RGB and tactile maps.
- ACT predicts 19-step action chunks.

Controlled Comparison

- T4–T6: tactile on/off × wrist camera on/off.
- Scene and egocentric RGB remain available in all conditions.

1,258 successful demonstrations · 1,000 physical evaluation rollouts



3. Results

Table 1. Vision-only vs. vision+tactile success for all 6 tasks

	Task	Wrist	V	V+T	Δ
T1	Tight-clearance block	on	15/30 (50.0%)	19/30 (63.3%)	+13.3 pp
T2a	Rigid block	on	15/25 (60.0%)	15/25 (60.0%)	0.0 pp
T2b	Sponge	on	19/25 (76.0%)	22/25 (88.0%)	+12.0 pp
T3	Straw insertion	on	26/60 (43.3%)	39/60 (65.0%)	+21.7 pp
T4	Opaque-bag retrieval	off	13/60 (21.7%)	49/60 (81.7%)	+60.0 pp
		on	30/60 (50.0%)	43/60 (71.7%)	+21.7 pp
T5	Plug insertion	off	9/60 (15.0%)	28/60 (46.7%)	+31.7 pp
		on	20/60 (33.3%)	30/60 (50.0%)	+16.7 pp
T6	Whiteboard wiping	off	13/60 (21.7%)	43/60 (71.7%)	+50.0 pp
		on	28/60 (46.7%)	39/60 (65.0%)	+18.3 pp

Tactile Gain without Wrist Images

+60.0 pp T4 retrieval

+31.7 pp T5 insertion

+50.0 pp T6 wiping

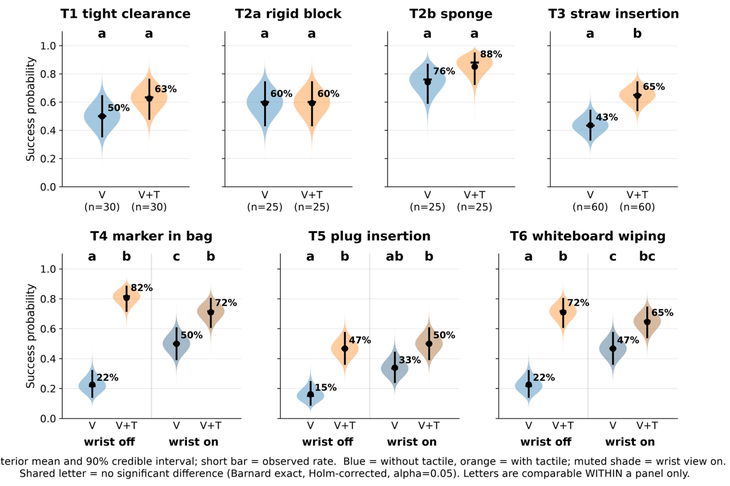
Without wrist images, V+T achieves higher success than wrist-equipped V on T4–T6 (significant on T4 and T6)

Four Sensor Conditions (T4–T6)

	Vision Only (V)	Vision + Touch (V+T)
Wrist OFF	V	V+T
Wrist ON	V	V+T

Scene and egocentric RGB are available in all conditions.

Task-Wise Success Across Sensor Conditions



Posterior mean and 90% credible interval; short bar = observed rate. Blue = without tactile, orange = with tactile; muted shade = wrist view on. Shared letter = no significant difference (Barnard exact, Holm-corrected, alpha=0.05). Letters are comparable WITHIN a panel only.

When Does Touch Help?

- **Smaller gains:** Visually observable placement tasks.
- **Larger gains:** Occluded retrieval, insertion, and sustained-contact manipulation.

Overlapping Information from Touch and Wrist Vision

- Smaller tactile gains with wrist vision are consistent with partially overlapping task-relevant information.
- Adding wrist vision does not consistently improve V+T success.

4. Conclusion and Future Work

Touch can substitute for wrist vision in the evaluated contact-rich manipulation tasks.

- Touch is especially useful when contact information is visually difficult to observe.
- V+T without wrist images outperforms wrist-equipped V on T4–T6.
- Touch complements scene-level vision rather than replacing it.

Future Work:

- Test multiple training seeds and checkpoint robustness.
- Evaluate additional robot platforms and tactile sensor designs.

Watch the robot in action
Scan to learn more about paper, project, and videos
https://annieyaj.github.io/tactile-vs-vision/
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